MoveIt 2 Quick Start

DR-xcobo 2.0 | v1.0 | ROS 2 Jazzy | English

Environment: Ubuntu 24.04 LTS Desktop with ROS 2 Jazzy Desktop installed. Use Bash. This package provides MoveIt 2 planning and mock execution in RViz2.

Download the ROS 2 MoveIt package ZIP into **~/Downloads/** and keep its filename. Geometry, joint structure and position limits are retained from the ROS 1 model; launch, planning and control configuration are adapted for ROS 2.

1. Install dependencies (once)

```
source /opt/ros/jazzy/setup.bash
sudo apt update
sudo apt install unzip build-essential python3-yaml \
  python3-colcon-common-extensions ros-jazzy-ament-cmake \
  ros-jazzy-ament-cmake-test \
  ros-jazzy-moveit ros-jazzy-ros2-control \
  ros-jazzy-ros2-controllers ros-jazzy-robot-state-publisher \
  ros-jazzy-rviz2
```

These commands assume ROS is installed. If the first line cannot find the file, install ROS 2 Jazzy first. Use a terminal that has not sourced ROS 1.

2. Extract & build (once)

```
mkdir -p ~/dr_xcobo_moveit_ws/src
cd ~/dr_xcobo_moveit_ws/src
unzip ~/Downloads/dr-xcobo-2.0-ros2-moveit-v1.0.zip
cd ~/dr_xcobo_moveit_ws
colcon build --symlink-install \
  --packages-select dr_xcobo_moveit_config
```

Keep the inner package name **dr_xcobo_moveit_config** and avoid duplicate copies in the workspace. The model and meshes are included; the separate URDF download is not required.

3. Source the workspace & launch

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_moveit_ws/install/setup.bash
ros2 launch dr_xcobo_moveit_config demo.launch.py
```

RViz2 opens after startup. The launcher uses **GenericSystem** mock hardware: it does not open a serial port, drive a physical robot, or simulate Gazebo physics. Continue on the next page.

Planning & Daily Use

DR-xcobo 2.0 | MoveIt 2 / RViz2 | English

4. Plan & execute in RViz2

1. Check settings: Fixed Frame is **base_link** and Planning Group is **arm**. Wait for a valid current robot state.

2. Set a goal: select **current** as the Start State. Drag the goal with the interactive marker, or adjust target joints in the Joints tab. Choose a reachable, collision-free goal.

3. Click Plan: inspect the preview. After a successful plan, click **Execute** to update the mock robot state. The controller moves the six arm joints; both fingers remain at their initial positions.

4. Stop: press **Ctrl+C** in the launch terminal. Mock execution checks the planning/control workflow, not physical performance or hardware collision safety.

Do I need to source each new terminal?

Yes. Repeat Step 3 on the previous page: load system ROS first, then this workspace, then launch. **source** affects only the current terminal; repeating it in the same configured terminal is unnecessary.

Optional: add the two source lines to **~/.bashrc**, but omit ros2 launch. Automatically source only one ROS version; do not mix Noetic or unrelated overlays into this environment.

Why is colcon build needed here?

This package uses ROS 2 **ament_cmake**. colcon build registers the package, model, configuration and launch files in the workspace install directory so ros2 launch can find them. Do not use catkin_make. Build initially and after adding files or changing build configuration; routine launches normally need no rebuild.

Execution fails or the package is not found

Confirm that the build succeeded and source again. Keep the launcher running; check controllers from a second terminal:

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_moveit_ws/install/setup.bash
ros2 control list_controllers
```

Both **joint_state_broadcaster** and **arm_controller** should be **active**. Otherwise, inspect the first error in the launch terminal. If planning fails, try a nearby, collision-free goal.

Model & mock limits

The original model has zero velocity/effort limits. Mock-only values are used here: arm 0.5 rad/s, 1.0 rad/s², 5 N·m; fingers 0.02 m/s, 0.04 m/s², 5 N, initially at 0.02 m. These are not hardware ratings and must not be used directly on a physical robot. See MIGRATION.md in the package.

Model, configuration, launch structure and official Jazzy documentation were checked. This package has not been built or run in a ROS environment.

Course resources | MoveIt 2 installation | Mock hardware reference

MoveIt 2 Quick Start

DR-xcobo 2.0 | v1.0 | ROS 2 Jazzy | 中文版

环境：Ubuntu 24.04 LTS 桌面版，已安装 ROS 2 Jazzy Desktop；使用 Bash 终端。此包用于 MoveIt 2 运动规划与 RViz2 模拟执行。

下载 ROS 2 MoveIt 功能包 ZIP，保留文件名并放在 ~/Downloads/。模型沿用原 ROS 1 包的几何、关节结构与位置限位；ROS 2 启动、规划和控制配置已重新适配。

1. 安装依赖（首次使用）

```
source /opt/ros/jazzy/setup.bash
sudo apt update
sudo apt install unzip build-essential python3-yaml \
  python3-colcon-common-extensions ros-jazzy-ament-cmake \
  ros-jazzy-ament-cmake-test \
  ros-jazzy-moveit ros-jazzy-ros2-control \
  ros-jazzy-ros2-controllers ros-jazzy-robot-state-publisher \
  ros-jazzy-rviz2
```

以上命令不安装完整 ROS；若第一行提示文件不存在，请先安装 ROS 2 Jazzy。不要在已加载 ROS 1 的终端中执行。

2. 解压并构建（首次使用）

```
mkdir -p ~/dr_xcobo_moveit_ws/src
cd ~/dr_xcobo_moveit_ws/src
unzip ~/Downloads/dr-xcobo-2.0-ros2-moveit-v1.0.zip
cd ~/dr_xcobo_moveit_ws
colcon build --symlink-install \
  --packages-select dr_xcobo_moveit_config
```

保留内部包名 dr_xcobo_moveit_config；避免在工作空间放入另一份同名包。此包自带模型和网格，无需再下载独立 URDF 包。

3. 加载工作空间并启动

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_moveit_ws/install/setup.bash
ros2 launch dr_xcobo_moveit_config demo.launch.py
```

启动后打开 RViz2。该入口仅使用模拟硬件 GenericSystem，不会打开串口或驱动真实机械臂；它也不是 Gazebo 物理仿真。下一页为规划操作。

规划与日常使用

DR-xcobo 2.0 | MoveIt 2 / RViz2 | 中文版

4. 在 RViz2 中规划与模拟执行

① **检查设置**：Fixed Frame 为 base_link，Planning Group 为 arm；等待机器人当前状态显示正常。

② **设置目标**：Start State 选择 current，使用交互标记拖动目标位姿，或在 Joints 页调整目标关节。选择可达且无碰撞的目标。

③ **点击 Plan**：先查看轨迹预览；规划成功后点击 Execute，观察模拟机器人状态更新。只有六个机械臂关节参与控制，两个夹指保持初始位置。

④ **退出**：在启动终端按 Ctrl+C。模拟执行用于检查规划与控制流程，不代表真实硬件性能或碰撞安全性。

新终端需要重新 source 吗?

需要。每个新终端都执行上一页第 3 步：先加载系统 ROS，再加载此工作空间，最后启动。source 只影响当前终端；已加载的同一终端无需反复执行。

可选：将第 3 步的两行 source 加入 ~/.bashrc，不要加入 ros2 launch。只自动加载一套 ROS 版本，避免与 Noetic 或其他工作空间混用。

为什么这里需要 colcon build?

本包使用 ROS 2 的 ament_cmake。colcon build 会把包、模型、配置和启动文件注册到工作空间 install 目录，供 ros2 launch 查找；不使用 catkin_make。首次使用、增加文件或修改构建配置后重新构建，日常启动通常无需重建。

无法执行或找不到功能包

先确认构建成功并重新 source。启动窗口保持运行，再开一个终端检查控制器：

```
source /opt/ros/jazzy/setup.bash
source ~/dr_xcobo_moveit_ws/install/setup.bash
ros2 control list_controllers
```

应看到 joint_state_broadcaster 和 arm_controller 都为 active。若不是，查看启动终端的首个错误；若规划失败，先换一个更近、无碰撞的目标。

模型与模拟参数

原模型的速度和力矩限位为 0。此版使用仅供模拟的值：机械臂 0.5 rad/s、1.0 rad/s²、5 N · m；夹指 0.02 m/s、0.04 m/s²、5 N，初始位置 0.02 m。这些不是设备额定参数，不可直接用于真实机器人。详见包内 MIGRATION.md。

已核对模型、配置、启动结构与官方 Jazzy 文档；尚未在 ROS 环境中构建或运行验证。

课程资源 | MoveIt 2 安装 | 模拟硬件说明